

Netflix CS

March 19, 2025

1 Netflix-business case study

Introduction

Netflix, Inc. is an American subscription streaming service and production company. It offers a library of films and television series through distribution deals as well as its own productions, known as Netflix Originals. As of March 31, 2023, with an estimated 232.5 million paid memberships in more than 190 countries, it is the most-subscribed video on demand streaming service.

Founded by Reed Hastings and Marc Randolph in Scotts Valley, California, Netflix initially operated as a DVD sales and rental business. However, within a year, it shifted its focus exclusively to DVD rentals. In 2007, the company introduced streaming media and video on demand services, marking a significant step in its evolution

Problem Statement:

Analysing the data and generate insights that could help Netflix in deciding which type of shows/movies to produce and how they can grow the business in different countries. 1.2 About Data This tabular dataset consists of data as of mid-2021, about 8807 movies and tv shows available on Netflix, along with details such as- cast, directors, ratings, release year, duration, etc. The data is available in a single csv file.

About Data:

This tabular dataset consists of data as of mid-2021, about 8807 movies and tv shows available on Netflix, along with details such as- cast, directors, ratings, release year, duration, etc. The data is available in a single csv file.

Features of the dataset:

The dataset provided to you consists of a list of all the TV shows/movies available on Netflix

Feature and its Description

Show ID-The ID of the show

Type Identifier- A Movie or TV Show

Title-Title of the Movie / Tv Show

Director-Director of the Movie

Cast-Actors involved in the movie/show

Country-Country where the movie/show was produced

Date_added-Date it was added on Netflix

Release_year-Actual Release year of the movie/show

Rating-TV Rating of the movie/show

Duration-Total Duration- in minutes or number of seasons

Listed_in-Genre

Feature-Description

Description-The summary description

Observations on the shape of data, data types of all the attributes, conversion of categorical attributes to 'category' (If required), missing value detection, statistical summary.

```
[1]: !pip install numpy pandas matplotlib seaborn
```

```
[2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import copy
import warnings
warnings.filterwarnings('ignore')
```

```
[3]: df=pd.read_csv("https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/
↪000/940/original/netflix.csv")
df.head(3)
```

```
[3]:  show_id    type      title      director \
0      s1  Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show      Blood & Water           NaN
2      s3  TV Show      Ganglands  Julien Leclercq

                                cast      country \
0                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...  South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...      NaN

    date_added  release_year  rating  duration \
0  September 25, 2021      2020  PG-13    90 min
1  September 24, 2021      2021  TV-MA    2 Seasons
2  September 24, 2021      2021  TV-MA    1 Season

                                listed_in \
0                                Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...
```

```

                                description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...

```

```
[4]: df.shape
```

```
[4]: (8807, 12)
```

```
[5]: df.dtypes
```

```

[5]: show_id      object
     type         object
     title        object
     director     object
     cast         object
     country      object
     date_added   object
     release_year  int64
     rating       object
     duration     object
     listed_in    object
     description  object
     dtype: object

```

```
[6]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   show_id         8807 non-null  object
 1   type            8807 non-null  object
 2   title           8807 non-null  object
 3   director        6173 non-null  object
 4   cast            7982 non-null  object
 5   country         7976 non-null  object
 6   date_added      8797 non-null  object
 7   release_year    8807 non-null  int64
 8   rating          8803 non-null  object
 9   duration        8804 non-null  object
10  listed_in       8807 non-null  object
11  description      8807 non-null  object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB

```

```
[7]: df.isnull().sum()
```

```
[7]: show_id      0
      type        0
      title       0
      director    2634
      cast        825
      country     831
      date_added  10
      release_year 0
      rating      4
      duration    3
      listed_in   0
      description 0
      dtype: int64
```

Calculating the count of null Values.

```
[8]: for i in df.columns:
      null_rate = df[i].isnull().sum()/df.shape[0] * 100
      if null_rate > 0:
          print(f"{i}'s null rate : {round(null_rate,2)}%")
```

```
director's null rate : 29.91%
cast's null rate : 9.37%
country's null rate : 9.44%
date_added's null rate : 0.11%
rating's null rate : 0.05%
duration's null rate : 0.03%
```

```
[9]: df.describe(include="object")
```

```
[9]:
```

	show_id	type	title	director	cast \
count	8807	8807	8807	6173	7982
unique	8807	2	8807	4528	7692
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough
freq	1	6131	1	19	19

	country	date_added	rating	duration \
count	7976	8797	8803	8804
unique	748	1767	17	220
top	United States	January 1, 2020	TV-MA	1 Season
freq	2818	109	3207	1793

	listed_in \
count	8807
unique	514
top	Dramas, International Movies
freq	362

	description
count	8807
unique	8775
top	Paranormal activity at a lush, abandoned prope...
freq	4

```
[10]: df.describe()
```

```
[10]:      release_year
count    8807.000000
mean     2014.180198
std        8.819312
min      1925.000000
25%      2013.000000
50%      2017.000000
75%      2019.000000
max      2021.000000
```

Insights: 1.Type of content- Among the 8807 items available on Netflix, 6131 of them are movies, accounting for nearly 70% of the total content.The remaining 30% consists of TV series.

2.Director- Rajiv Chilaka holds the top position on the director list, with 19 credits to his name. He specializes in creating animated movies for children.

3.Cast- David Attenborough leads the actor list with 19 appearances in various films and shows on Netflix.

4.Country- The USA ranks at the top as the country with the highest production contribution to Netflix, accounting for 35% of the total content.

5.Date Added- January 1, 2020, stands out as the peak date for content uploads on Netflix. On that day alone, approximately 109 different shows and movies were added to the platform.

6.Ratings- There are 17 different types of ratings present on Netflix. The “TV-MA” (Mature Audience Only) rating dominates the charts, covering almost 36% of the total shows and movies on the platform with this rating.

7.Listed_in- Dramas and International Movies are more in Netflix.

1.0.1 Duplicate Record Detection

```
[11]: df.duplicated().value_counts()
```

```
[11]: False    8807
      Name: count, dtype: int64
```

No duplicate records found in Netflix dataset

1.0.2 Value counts and unique attributes

```
[12]: for i in df.columns:
      print(i,":",df[i].nunique())
      print("-"*25)
```

```
show_id : 8807
-----
type : 2
-----
title : 8807
-----
director : 4528
-----
cast : 7692
-----
country : 748
-----
date_added : 1767
-----
release_year : 74
-----
rating : 17
-----
duration : 220
-----
listed_in : 514
-----
description : 8775
-----
```

Insights:

1. ShowId is unique hence it has 8807 unique Id's are present.
2. Netflix two types content movies and TVshows hence type has 2 unique values
3. Every title is unique hence 8807 titles are present
4. 4528 unique directors are present, including movies and TVshows directors
5. Netflix has running streaming in 748 countries

```
[13]: columns=['type', 'director', 'country', 'date_added',
      'release_year', 'rating', 'duration', 'listed_in']
      for i in columns:
          print(i,":")
          print(df[i].value_counts().head(5))
          print("-"*20)
```

```
type :
type
Movie      6131
```

TV Show 2676
Name: count, dtype: int64

director :
director
Rajiv Chilaka 19
Raúl Campos, Jan Suter 18
Suhas Kadav 16
Marcus Raboy 16
Jay Karas 14
Name: count, dtype: int64

country :
country
United States 2818
India 972
United Kingdom 419
Japan 245
South Korea 199
Name: count, dtype: int64

date_added :
date_added
January 1, 2020 109
November 1, 2019 89
March 1, 2018 75
December 31, 2019 74
October 1, 2018 71
Name: count, dtype: int64

release_year :
release_year
2018 1147
2017 1032
2019 1030
2020 953
2016 902
Name: count, dtype: int64

rating :
rating
TV-MA 3207
TV-14 2160
TV-PG 863
R 799
PG-13 490
Name: count, dtype: int64

```

duration :
duration
1 Season      1793
2 Seasons     425
3 Seasons     199
90 min        152
97 min        146
Name: count, dtype: int64
-----
listed_in :
listed_in
Dramas, International Movies      362
Documentaries                    359
Stand-Up Comedy                  334
Comedies, Dramas, International Movies  274
Dramas, Independent Movies, International Movies  252
Name: count, dtype: int64
-----

```

Above result is showing top 5 categories in all features

Handling Null Values

```

[14]: df['rating'].replace({'74 min': np.nan, '84 min': np.nan, '66 min': np.nan},
    ↪ inplace=True)
df['rating'].isnull().sum()

```

```

[14]: np.int64(7)

```

The rating column has a total of 7 Null Values

```

[15]: df[df['rating'].isnull()]

```

```

[15]:
   show_id  type  title \
5541  s5542  Movie      Louis C.K. 2017
5794  s5795  Movie      Louis C.K.: Hilarious
5813  s5814  Movie      Louis C.K.: Live at the Comedy Store
5989  s5990  Movie  13TH: A Conversation with Oprah Winfrey & Ava ...
6827  s6828  TV Show      Gargantia on the Verdurous Planet
7312  s7313  TV Show      Little Lunch
7537  s7538  Movie      My Honor Was Loyalty

   director  cast \
5541  Louis C.K.      Louis C.K.
5794  Louis C.K.      Louis C.K.
5813  Louis C.K.      Louis C.K.
5989      NaN      Oprah Winfrey, Ava DuVernay
6827      NaN  Kaito Ishikawa, Hisako Kanemoto, Ai Kayano, Ka...
7312      NaN  Flynn Curry, Olivia Deeble, Madison Lu, Oisín ...

```

7537 Alessandro Pepe Leone Frisa, Paolo Vaccarino, Francesco Miglio...

	country	date_added	release_year	rating	duration	\
5541	United States	April 4, 2017	2017	NaN	NaN	
5794	United States	September 16, 2016	2010	NaN	NaN	
5813	United States	August 15, 2016	2015	NaN	NaN	
5989	NaN	January 26, 2017	2017	NaN	37 min	
6827	Japan	December 1, 2016	2013	NaN	1 Season	
7312	Australia	February 1, 2018	2015	NaN	1 Season	
7537	Italy	March 1, 2017	2015	NaN	115 min	

	listed_in	\
5541	Movies	
5794	Movies	
5813	Movies	
5989	Movies	
6827	Anime Series, International TV Shows	
7312	Kids' TV, TV Comedies	
7537	Dramas	

	description
5541	Louis C.K. muses on religion, eternal love, gi...
5794	Emmy-winning comedy writer Louis C.K. brings h...
5813	The comic puts his trademark hilarious/thought...
5989	Oprah Winfrey sits down with director Ava DuVe...
6827	After falling through a wormhole, a space-dwel...
7312	Adopting a child's perspective, this show take...
7537	Amid the chaos and horror of World War II, a c...

```
[16]: #using google to find the rating for 7 titles in Netflix
ratings_dict = {
    "13TH: A Conversation with Oprah Winfrey & Ava DuVernay": 'TV-PG',
    "Gargantia on the Verdurous Planet": 'TV-PG',
    "Little Lunch": 'TV-Y',
    "My Honor Was Loyalty": 'PG-13',
    "Louis C.K. 2017": 'TV-MA',
    "Louis C.K.: Hilarious": 'TV-MA',
    "Louis C.K.: Live at the Comedy Store": 'TV-MA'
}
df['rating'] = df['title'].map(ratings_dict).fillna(df['rating'])
```

```
[17]: df['rating'].isnull().sum()
```

```
[17]: np.int64(0)
```

```
[18]: # for the missing 3 titles got the correct duration from google
durn = {"Louis C.K. 2017" : '74 min',
```

```
"Louis C.K.: Hilarious" : '83 min',
"Louis C.K.: Live at the Comedy Store":'65 min'}
df['duration'] = df['title'].map(durn).fillna(df['duration'])
```

```
[19]: #since there are only 10 dates added values missing , searched and found from
      ↪google
date = {"A Young Doctor's Notebook and Other Stories" : 'September 15, 2014',
"Anthony Bourdain: Parts Unknown" : 'June 1, 2020',
"Frasier" : 'April 1, 2015',
"Friends" : 'January 1, 2015',
"Gunslinger Girl" : 'January 4, 2016',
"Kikoriki" : 'July 6, 2017',
"La Familia P. Luche" : 'June 1, 2015',
"Maron" : 'November 2, 2018',"Red vs. Blue" : 'March 21, 2015',
"The Adventures of Figaro Pho": 'January 1, 2016'}
df['date_added'] = df['title'].map(date).fillna(df['date_added'])
```

```
[20]: #rechecking the null values in columns
df.isnull().sum()
```

```
[20]: show_id      0
      type        0
      title       0
      director    2634
      cast        825
      country     831
      date_added  0
      release_year 0
      rating      0
      duration    0
      listed_in   0
      description 0
      dtype: int64
```

```
[21]: # filling the na values in country column with the mode value
df['country'] = df['country'].fillna(df['country'].mode()[0])
```

```
[22]: # filling director and cast column with unkown
df['director'].fillna('Unknown Director',inplace = True)
df['cast'].fillna('Unknown cast',inplace = True)
```

```
[23]: df.isnull().sum()
```

```
[23]: show_id      0
      type        0
      title       0
      director    0
```

```

cast          0
country       0
date_added    0
release_year  0
rating        0
duration      0
listed_in     0
description    0
dtype: int64

```

```

[24]: # Convert to string first, then strip spaces
df['date_added'] = df['date_added'].astype(str).strip()
# Convert 'date_added' to datetime
df['date_added'] = pd.to_datetime(df['date_added'], format='%B %d, %Y',
    ↪errors='coerce')
# Adding new columns to explore data separately by year, month, and week
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month_name()
df['week_added'] = df['date_added'].dt.isocalendar().week
df.head()

```

```

[24]: show_id    type    title    director \
0      s1      Movie    Dick Johnson Is Dead    Kirsten Johnson
1      s2  TV Show      Blood & Water    Unknown Director
2      s3  TV Show      Ganglands    Julien Leclercq
3      s4  TV Show    Jailbirds New Orleans    Unknown Director
4      s5  TV Show      Kota Factory    Unknown Director

                                cast    country \
0                                Unknown cast    United States
1    Ama Qamata, Khosi Ngema, Gail Mabalan...    South Africa
2    Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...    United States
3                                Unknown cast    United States
4    Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...    India

    date_added  release_year  rating  duration \
0  2021-09-25          2020  PG-13    90 min
1  2021-09-24          2021  TV-MA  2 Seasons
2  2021-09-24          2021  TV-MA  1 Season
3  2021-09-24          2021  TV-MA  1 Season
4  2021-09-24          2021  TV-MA  2 Seasons

                                listed_in \
0                                Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2    Crime TV Shows, International TV Shows, TV Act...
3                                Docuseries, Reality TV

```

4 International TV Shows, Romantic TV Shows, TV ...

	description	year_added	month_added	\
0	As her father nears the end of his life, filmm...	2021	September	
1	After crossing paths at a party, a Cape Town t...	2021	September	
2	To protect his family from a powerful drug lor...	2021	September	
3	Feuds, flirtations and toilet talk go down amo...	2021	September	
4	In a city of coaching centers known to train I...	2021	September	

	week_added
0	38
1	38
2	38
3	38
4	38

```
[85]: #Creating a backup of the original dataframe
df1 = df.copy(deep = True)
#saving the file as backup intermediate stage.
df.to_csv('netflix1.csv')
```

```
[86]: #It is difficult to analyse the values in director,cast,country because it is
      ↪separated by comma
#For better analyses exploding them to separate rows
df1["director"] = df["director"].str.split(", ")
df1["cast"] = df["cast"].str.split(", ")
df1["country"] = df["country"].str.split(", ")
df1 = df1.explode(['director'])
df1 = df1.explode(['cast'])
df1 = df1.explode(['country'])
df1.head(3)
```

```
[86]: show_id    type    title    director    cast \
0      s1      Movie  Dick Johnson Is Dead  Kirsten Johnson  Unknown cast
1      s2      TV Show    Blood & Water  Unknown Director    Ama Qamata
1      s2      TV Show    Blood & Water  Unknown Director    Khosi Ngema
```

	country	date_added	release_year	rating	duration	\
0	United States	2021-09-25	2020	PG-13	90 min	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	

	listed_in	\
0	Documentaries	
1	International TV Shows, TV Dramas, TV Mysteries	
1	International TV Shows, TV Dramas, TV Mysteries	

		description	year_added	month_added	\
0		As her father nears the end of his life, filmm...	2021	September	
1		After crossing paths at a party, a Cape Town t...	2021	September	
1		After crossing paths at a party, a Cape Town t...	2021	September	

	week_added
0	38
1	38
1	38

Checking the columns for Null Values

```
[87]: df1['country'].isna().sum()
```

```
[87]: np.int64(0)
```

```
[88]: df1['duration_cpy']=df1['duration'].str.replace(" min","")
df1.head()
```

```
[88]:
```

	show_id	type	title	director	cast	\
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown cast	
1	s2	TV Show	Blood & Water	Unknown Director	Ama Qamata	
1	s2	TV Show	Blood & Water	Unknown Director	Khosi Ngema	
1	s2	TV Show	Blood & Water	Unknown Director	Gail Mabalane	
1	s2	TV Show	Blood & Water	Unknown Director	Thabang Molaba	

	country	date_added	release_year	rating	duration	\
0	United States	2021-09-25	2020	PG-13	90 min	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	

	listed_in	\
0	Documentaries	
1	International TV Shows, TV Dramas, TV Mysteries	
1	International TV Shows, TV Dramas, TV Mysteries	
1	International TV Shows, TV Dramas, TV Mysteries	
1	International TV Shows, TV Dramas, TV Mysteries	

		description	year_added	month_added	\
0		As her father nears the end of his life, filmm...	2021	September	
1		After crossing paths at a party, a Cape Town t...	2021	September	
1		After crossing paths at a party, a Cape Town t...	2021	September	
1		After crossing paths at a party, a Cape Town t...	2021	September	
1		After crossing paths at a party, a Cape Town t...	2021	September	

	week_added	duration_cpy
--	------------	--------------

0	38	90
1	38	2 Seasons
1	38	2 Seasons
1	38	2 Seasons
1	38	2 Seasons

```
[89]: #df1['duration_cpy']=df1['duration_cpy'].copy()
df1['duration_cpy']=df1['duration_cpy'].str.replace(" Seasons","")
df1['duration_cpy']=df1['duration_cpy'].str.replace(" Season","")
df1['duration_cpy']=df1['duration_cpy'].astype('int')
df1.head()
```

```
[89]: show_id    type    title    director    cast \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson  Unknown cast
1      s2  TV Show    Blood & Water  Unknown Director    Ama Qamata
1      s2  TV Show    Blood & Water  Unknown Director    Khosi Ngema
1      s2  TV Show    Blood & Water  Unknown Director    Gail Mabalane
1      s2  TV Show    Blood & Water  Unknown Director    Thabang Molaba
```

	country	date_added	release_year	rating	duration	\
0	United States	2021-09-25	2020	PG-13	90 min	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	
1	South Africa	2021-09-24	2021	TV-MA	2 Seasons	

	listed_in	\
0	Documentaries	
1	International TV Shows, TV Dramas, TV Mysteries	
1	International TV Shows, TV Dramas, TV Mysteries	
1	International TV Shows, TV Dramas, TV Mysteries	
1	International TV Shows, TV Dramas, TV Mysteries	

	description	year_added	month_added	\
0	As her father nears the end of his life, filmm...	2021	September	
1	After crossing paths at a party, a Cape Town t...	2021	September	
1	After crossing paths at a party, a Cape Town t...	2021	September	
1	After crossing paths at a party, a Cape Town t...	2021	September	
1	After crossing paths at a party, a Cape Town t...	2021	September	

	week_added	duration_cpy
0	38	90
1	38	2
1	38	2
1	38	2
1	38	2

```
[90]: df1['duration_cpy'].unique()
```

```
[90]: array([ 90,  2,  1,  91, 125,  9, 104, 127,  4,  67,  94,  5, 161,
          61, 166, 147, 103,  97, 106, 111,  3, 110, 105,  96, 124, 116,
          98,  23, 115, 122,  99,  88, 100,  6, 102,  93,  95,  85,  83,
          113,  13, 182,  48, 145,  87,  92,  80, 117, 128, 119, 143, 114,
          118, 108,  63, 121, 142, 154, 120,  82, 109, 101,  86, 229,  76,
          89, 156, 112, 107, 129, 135, 136, 165, 150, 133,  70,  84, 140,
          78,  7,  64,  59, 139,  69, 148, 189, 141, 130, 138,  81, 132,
          10, 123,  65,  68,  66,  62,  74, 131,  39,  46,  38,  8,  17,
          126, 155, 159, 137,  12, 273,  36,  34,  77,  60,  49,  58,  72,
          204, 212,  25,  73,  29,  47,  32,  35,  71, 149,  33,  15,  54,
          224, 162,  37,  75,  79,  55, 158, 164, 173, 181, 185,  21,  24,
          51, 151,  42,  22, 134, 177,  52,  14,  53,  57,  28,  50,  26,
          45, 171,  27,  44, 146,  20, 157, 203,  41,  30, 194, 233, 237,
          230, 195, 253, 152, 190, 160, 208, 180, 144, 174, 170, 192, 209,
          187, 172,  16, 186,  11, 193, 176,  56, 169,  40, 168, 312, 153,
          214,  31, 163,  19, 179,  43, 200, 196, 167, 178, 228,  18, 205,
          201, 191])
```

```
[91]: df1['director'].isna().sum()
```

```
[91]: np.int64(0)
```

```
[92]: df1['cast'].isna().sum()
```

```
[92]: np.int64(0)
```

```
[93]: df1.shape
```

```
[93]: (89382, 16)
```

```
[94]: #TVShows VS Rating, and Movies VS Rating
df1.groupby(['type', 'rating'])['show_id'].count()
```

```
[94]: type      rating
Movie      G          878
          NC-17         55
          NR          624
          PG         5182
          PG-13        7821
          R         12841
          TV-14       11515
          TV-G          842
          TV-MA       18205
          TV-PG        4216
          TV-Y        1461
          TV-Y7       1643
```

	TV-Y7-FV	31
	UR	32
TV Show	NR	52
	R	24
	TV-14	5922
	TV-G	487
	TV-MA	11658
	TV-PG	2164
	TV-Y	1291
	TV-Y7	2430
	TV-Y7-FV	8

Name: show_id, dtype: int64

```
[95]: df1.groupby(df1['type'])['show_id'].nunique()
```

```
[95]: type
Movie      6131
TV Show    2676
Name: show_id, dtype: int64
```

```
[96]: df1['country'].unique()
```

```
[96]: array(['United States', 'South Africa', 'India', 'Ghana', 'Burkina Faso',
'United Kingdom', 'Germany', 'Ethiopia', 'Czech Republic',
'Mexico', 'Turkey', 'Australia', 'France', 'Finland', 'China',
'Canada', 'Japan', 'Nigeria', 'Spain', 'Belgium', 'South Korea',
'Singapore', 'Italy', 'Romania', 'Argentina', 'Venezuela',
'Hong Kong', 'Russia', '', 'Ireland', 'Nepal', 'New Zealand',
'Brazil', 'Greece', 'Jordan', 'Colombia', 'Switzerland', 'Israel',
'Taiwan', 'Bulgaria', 'Algeria', 'Poland', 'Saudi Arabia',
'Thailand', 'Indonesia', 'Egypt', 'Denmark', 'Kuwait',
'Netherlands', 'Malaysia', 'Vietnam', 'Hungary', 'Sweden',
'Lebanon', 'Syria', 'Philippines', 'Iceland',
'United Arab Emirates', 'Norway', 'Qatar', 'Mauritius', 'Austria',
'Cameroon', 'Palestine', 'Uruguay', 'United Kingdom,', 'Kenya',
'Chile', 'Luxembourg', 'Cambodia', 'Bangladesh', 'Portugal',
'Cayman Islands', 'Senegal', 'Serbia', 'Malta', 'Namibia',
'Angola', 'Peru', 'Mozambique', 'Cambodia,', 'Belarus', 'Zimbabwe',
'Puerto Rico', 'Pakistan', 'Cyprus', 'Guatemala', 'Iraq', 'Malawi',
'Paraguay', 'Croatia', 'Iran', 'West Germany', 'United States,',
'Albania', 'Georgia', 'Soviet Union', 'Morocco', 'Slovakia',
'Ukraine', 'Bermuda', 'Ecuador', 'Armenia', 'Mongolia', 'Bahamas',
'Sri Lanka', 'Latvia', 'Liechtenstein', 'Cuba', 'Nicaragua',
'Poland,', 'Slovenia', 'Dominican Republic', 'Samoa', 'Azerbaijan',
'Botswana', 'Vatican City', 'Jamaica', 'Kazakhstan', 'Lithuania',
'Afghanistan', 'Somalia', 'Sudan', 'Panama', 'Uganda',
'East Germany', 'Montenegro'], dtype=object)
```

```
[97]: #replacing empty cell with mode value
# The mode of the 'country' column
country_mode = df['country'].mode()[0]
# Use loc to select and replace empty strings in the 'country' column
df1.loc[df1['country'] == "", 'country'] = country_mode
```

```
[98]: df1['country'].unique()
```

```
[98]: array(['United States', 'South Africa', 'India', 'Ghana', 'Burkina Faso',
'United Kingdom', 'Germany', 'Ethiopia', 'Czech Republic',
'Mexico', 'Turkey', 'Australia', 'France', 'Finland', 'China',
'Canada', 'Japan', 'Nigeria', 'Spain', 'Belgium', 'South Korea',
'Singapore', 'Italy', 'Romania', 'Argentina', 'Venezuela',
'Hong Kong', 'Russia', 'Ireland', 'Nepal', 'New Zealand', 'Brazil',
'Greece', 'Jordan', 'Colombia', 'Switzerland', 'Israel', 'Taiwan',
'Bulgaria', 'Algeria', 'Poland', 'Saudi Arabia', 'Thailand',
'Indonesia', 'Egypt', 'Denmark', 'Kuwait', 'Netherlands',
'Malaysia', 'Vietnam', 'Hungary', 'Sweden', 'Lebanon', 'Syria',
'Philippines', 'Iceland', 'United Arab Emirates', 'Norway',
'Qatar', 'Mauritius', 'Austria', 'Cameroon', 'Palestine',
'Uruguay', 'United Kingdom,', 'Kenya', 'Chile', 'Luxembourg',
'Cambodia', 'Bangladesh', 'Portugal', 'Cayman Islands', 'Senegal',
'Serbia', 'Malta', 'Namibia', 'Angola', 'Peru', 'Mozambique',
'Cambodia,', 'Belarus', 'Zimbabwe', 'Puerto Rico', 'Pakistan',
'Cyprus', 'Guatemala', 'Iraq', 'Malawi', 'Paraguay', 'Croatia',
'Iran', 'West Germany', 'United States,', 'Albania', 'Georgia',
'Soviet Union', 'Morocco', 'Slovakia', 'Ukraine', 'Bermuda',
'Ecuador', 'Armenia', 'Mongolia', 'Bahamas', 'Sri Lanka', 'Latvia',
'Liechtenstein', 'Cuba', 'Nicaragua', 'Poland,', 'Slovenia',
'Dominican Republic', 'Samoa', 'Azerbaijan', 'Botswana',
'Vatican City', 'Jamaica', 'Kazakhstan', 'Lithuania',
'Afghanistan', 'Somalia', 'Sudan', 'Panama', 'Uganda',
'East Germany', 'Montenegro'], dtype=object)
```

```
[99]: df1['cast'].unique()
```

```
[99]: array(['Unknown cast', 'Ama Qamata', 'Khosi Ngema', ..., 'Malkeet Rauni',
'Anita Shabdish', 'Chittaranjan Tripathy'],
shape=(36440,), dtype=object)
```

```
[100]: df1['director'].nunique()
```

```
[100]: 4994
```

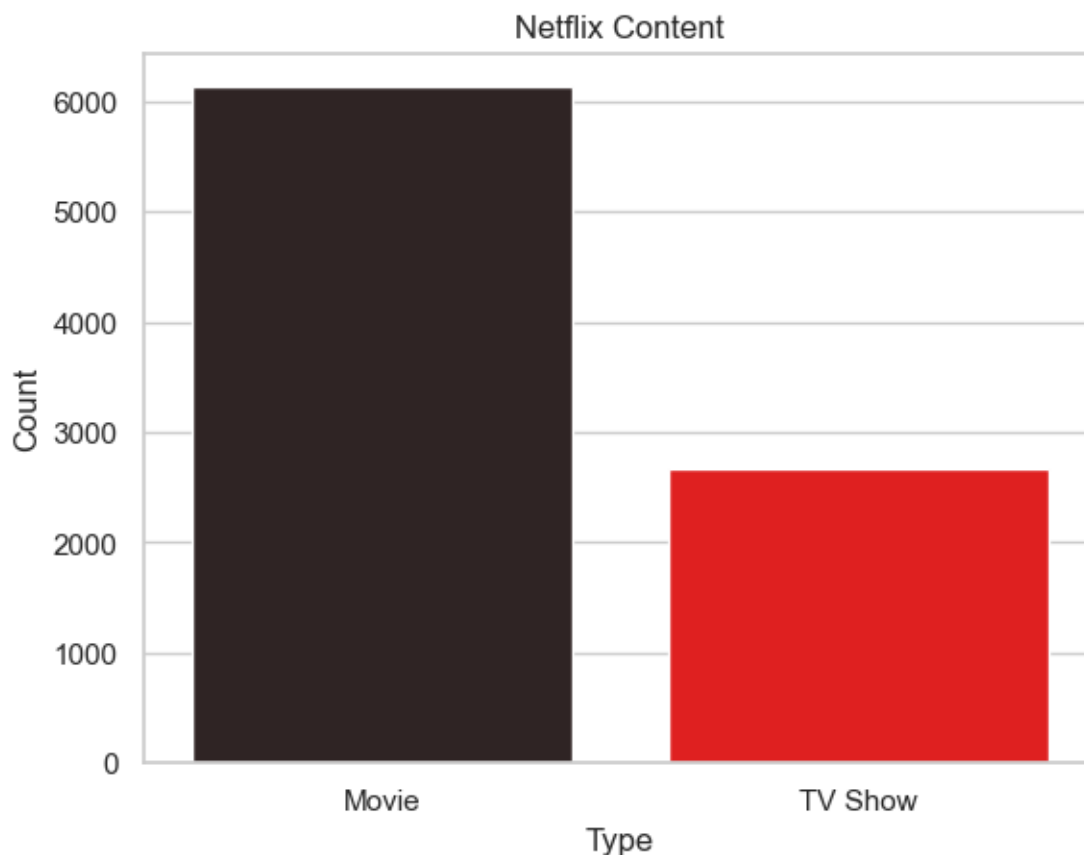
```
[101]: #saving the file as final cleaned for backup
df1.to_csv('netflix_cleaned.csv')
```

```
[102]: #1.6 Graphical and Non Graphical Analysis
```

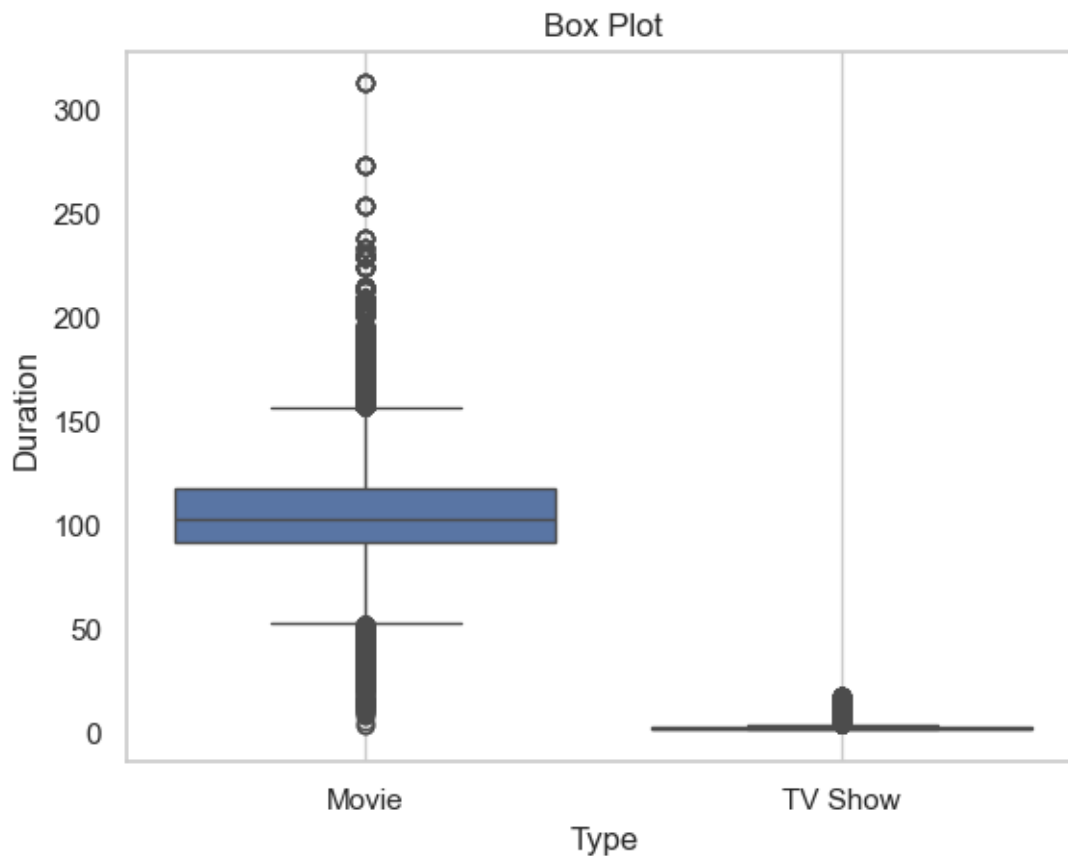
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1.1 Graphical and Non Graphical Analysis

```
[103]: # Group by 'type' and count unique 'show_id'
type_counts = df1.groupby('type')['show_id'].nunique().reset_index()
type_counts.columns = ['type', 'unique_show_id_count']
# Create a count plot
sns.barplot(x='type', y='unique_show_id_count', data=type_counts,
            hue='type', color="red")
plt.title('Netflix Content')
plt.xlabel('Type')
plt.ylabel('Count')
plt.show()
```

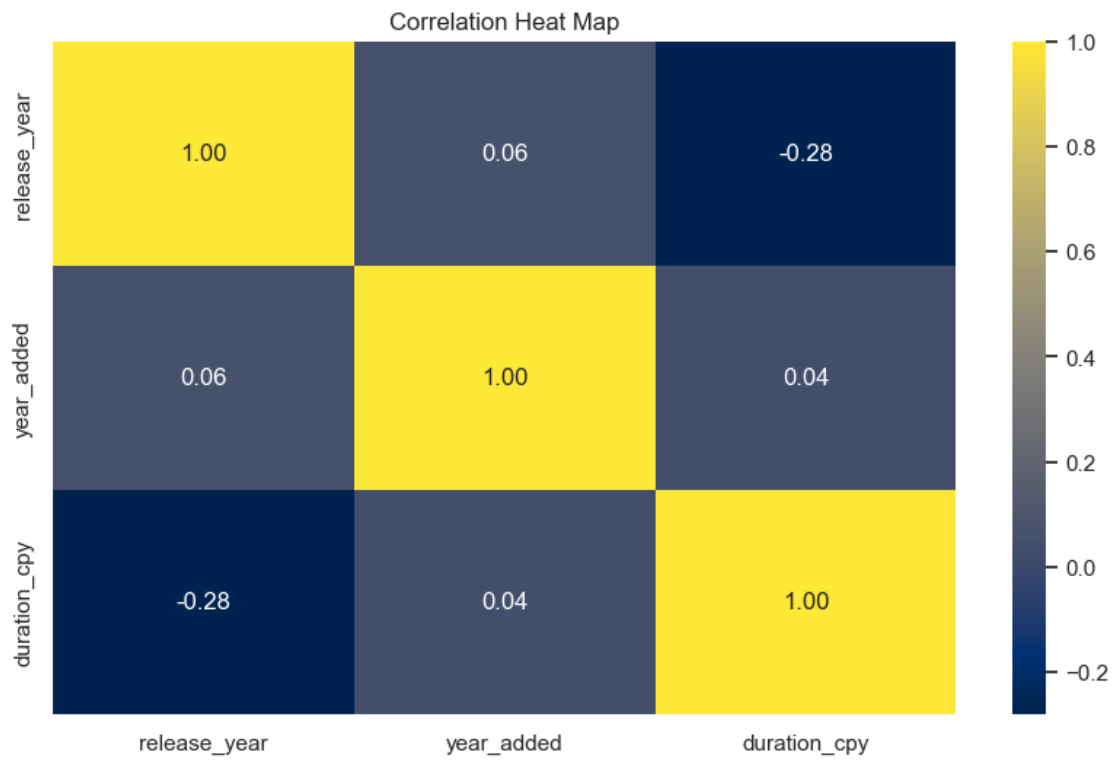


```
[104]: #box plot
sns.boxplot(data = df1,x= 'type',y = 'duration_cpy')
plt.title('Box Plot')
plt.grid()
plt.xlabel('Type')
plt.ylabel('Duration')
plt.show()
```

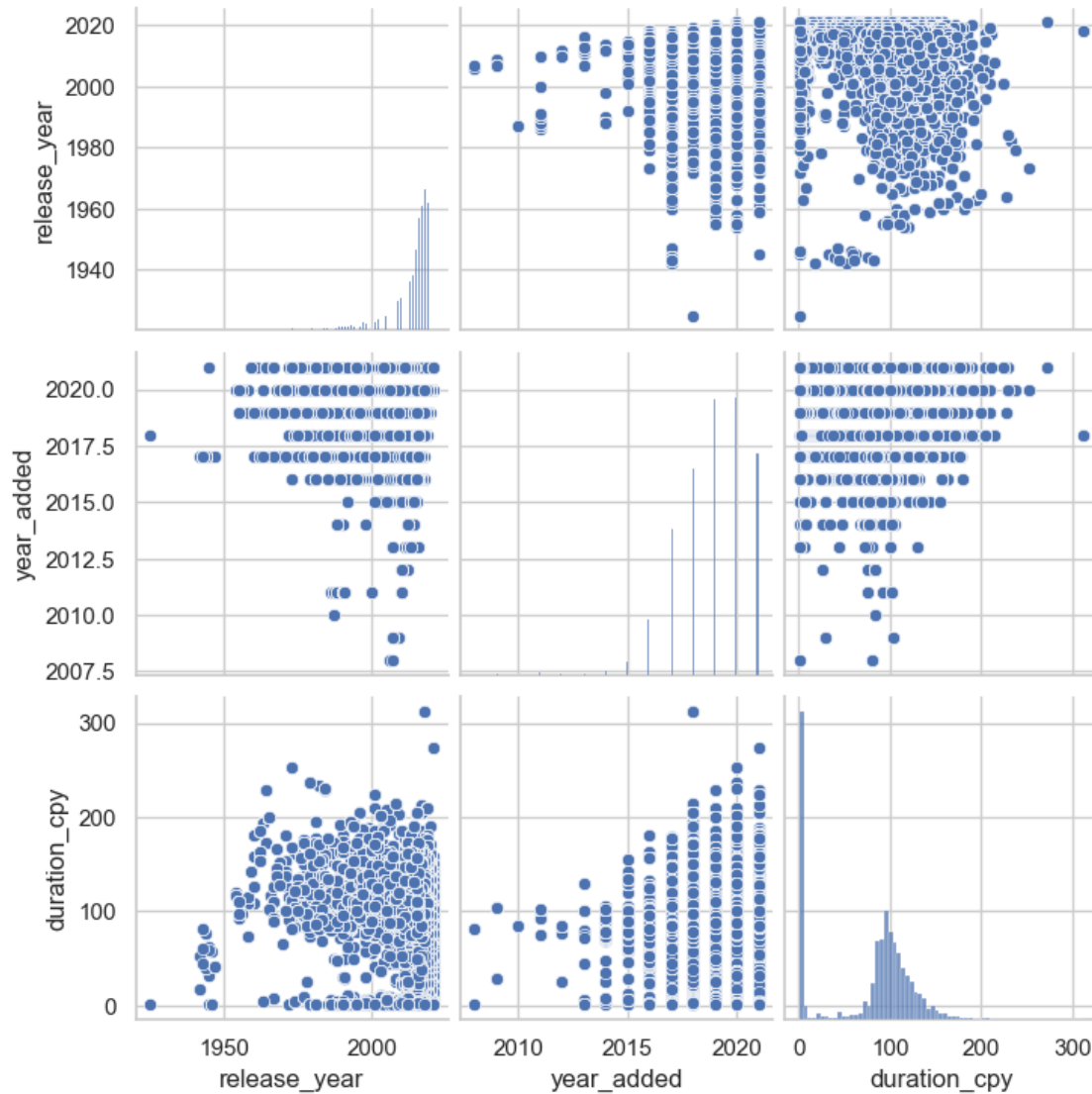


```
[105]: numerical_columns = ['release_year', 'year_added', 'duration_cpy']
correlation_matrix = df1[numerical_columns].corr()
```

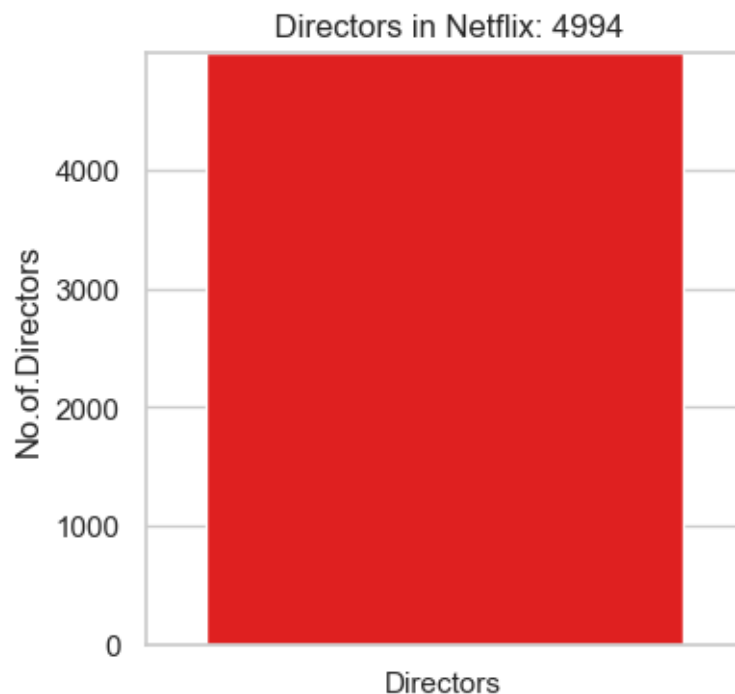
```
[106]: plt.figure(figsize=(10, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='cividis', fmt='.2f')
plt.title('Correlation Heat Map')
plt.show()
```



```
[107]: sns.pairplot(df1[numerical_columns])  
plt.grid()  
plt.show()
```



```
[108]: num_directors = df1['director'].nunique()
# Create a bar plot
plt.figure(figsize=(4, 4))
sns.barplot(x=['Directors'], y=[num_directors], color = 'red')
plt.ylabel('No.of.Directors')
plt.title(f' Directors in Netflix: {num_directors}')
plt.ylim(0, num_directors + 1)
plt.show()
```



```
[109]: df1.head()
```

```
[109]:  show_id    type      title      director      cast \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson  Unknown cast
1      s2  TV Show    Blood & Water  Unknown Director    Ama Qamata
1      s2  TV Show    Blood & Water  Unknown Director    Khosi Ngema
1      s2  TV Show    Blood & Water  Unknown Director    Gail Mabalane
1      s2  TV Show    Blood & Water  Unknown Director    Thabang Molaba

      country date_added  release_year  rating  duration \
0  United States 2021-09-25         2020  PG-13    90 min
1  South Africa 2021-09-24         2021  TV-MA  2 Seasons
1  South Africa 2021-09-24         2021  TV-MA  2 Seasons
1  South Africa 2021-09-24         2021  TV-MA  2 Seasons
1  South Africa 2021-09-24         2021  TV-MA  2 Seasons

      listed_in \
0      Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
1  International TV Shows, TV Dramas, TV Mysteries
1  International TV Shows, TV Dramas, TV Mysteries
1  International TV Shows, TV Dramas, TV Mysteries
```

```
description  year_added  month_added \
```

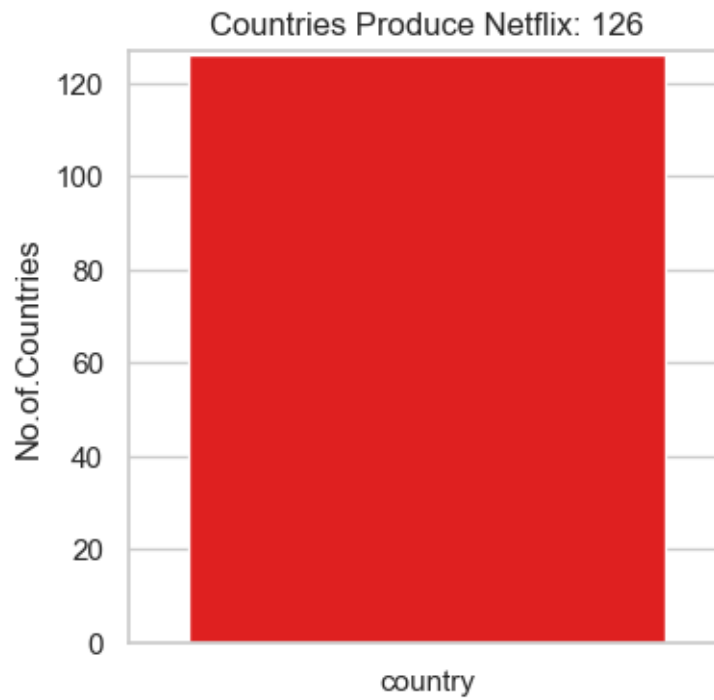
0	As her father nears the end of his life, filmm...	2021	September
1	After crossing paths at a party, a Cape Town t...	2021	September
1	After crossing paths at a party, a Cape Town t...	2021	September
1	After crossing paths at a party, a Cape Town t...	2021	September
1	After crossing paths at a party, a Cape Town t...	2021	September

	week_added	duration_cpy
0	38	90
1	38	2
1	38	2
1	38	2
1	38	2

```
[110]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   show_id               8807 non-null   object
1   type                  8807 non-null   object
2   title                 8807 non-null   object
3   director              8807 non-null   object
4   cast                  8807 non-null   object
5   country               8807 non-null   object
6   date_added            8807 non-null   datetime64[ns]
7   release_year          8807 non-null   int64
8   rating                8807 non-null   object
9   duration              8807 non-null   object
10  listed_in             8807 non-null   object
11  description            8807 non-null   object
12  year_added            8807 non-null   int32
13  month_added           8807 non-null   object
14  week_added            8807 non-null   UInt32
dtypes: UInt32(1), datetime64[ns](1), int32(1), int64(1), object(11)
memory usage: 972.0+ KB
```

```
[111]: num_countries = df1['country'].nunique()
plt.figure(figsize=(4, 4))
sns.barplot(x=['country'], y=[num_countries],color = 'red')
plt.ylabel('No.of.Countries')
plt.title(f'Countries Produce Netflix: {num_countries}')
plt.ylim(0, num_countries + 1)
plt.show()
```



```
[112]: # Number of MOvies,TV shows added yearwise
grouped_data = df1.groupby(['type', 'year_added']).size().
    ↪reset_index(name='country')
grouped_data
```

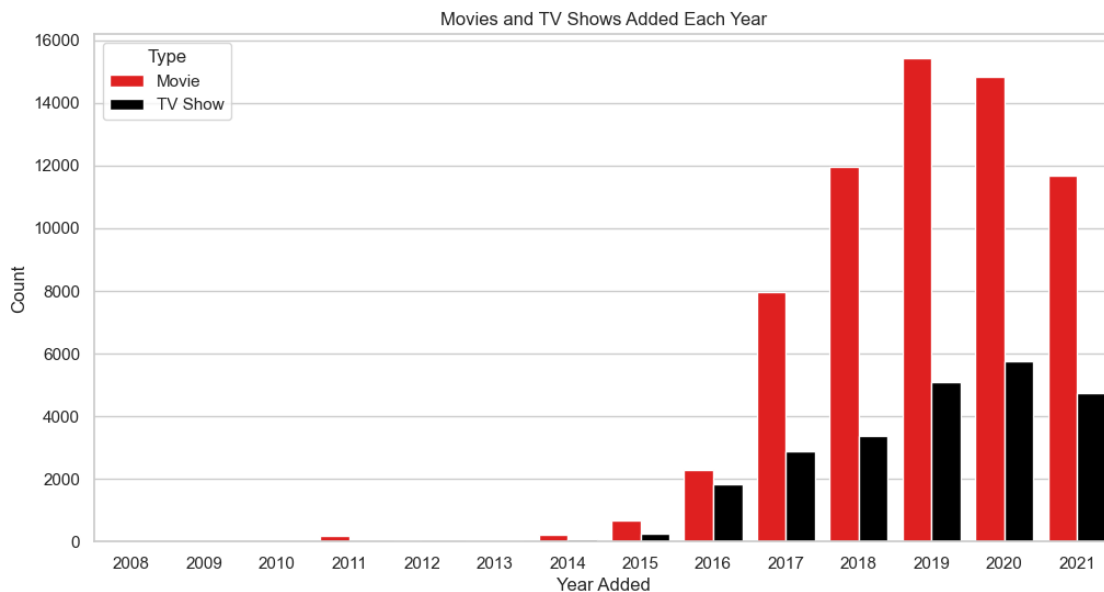
```
[112]:
```

	type	year_added	country
0	Movie	2008	6
1	Movie	2009	18
2	Movie	2010	10
3	Movie	2011	200
4	Movie	2012	27
5	Movie	2013	33
6	Movie	2014	223
7	Movie	2015	675
8	Movie	2016	2295
9	Movie	2017	7970
10	Movie	2018	11950
11	Movie	2019	15420
12	Movie	2020	14844
13	Movie	2021	11675
14	TV Show	2008	1
15	TV Show	2013	48
16	TV Show	2014	75
17	TV Show	2015	268

18	TV Show	2016	1814
19	TV Show	2017	2879
20	TV Show	2018	3370
21	TV Show	2019	5100
22	TV Show	2020	5755
23	TV Show	2021	4726

```
[113]: # Group by type and year, count occurrences
grouped_data = df1.groupby(['type', 'year_added']).size().
    ↪reset_index(name='count')
# Define color palette
palette = ['red', 'black']
# Create a bar plot
plt.figure(figsize=(12, 6))
sns.barplot(data=grouped_data, x='year_added', y='count', hue='type',
    ↪palette=palette)

# Labels and title
plt.xlabel('Year Added')
plt.ylabel('Count')
plt.title('Movies and TV Shows Added Each Year')
plt.legend(title='Type')
plt.show()
```

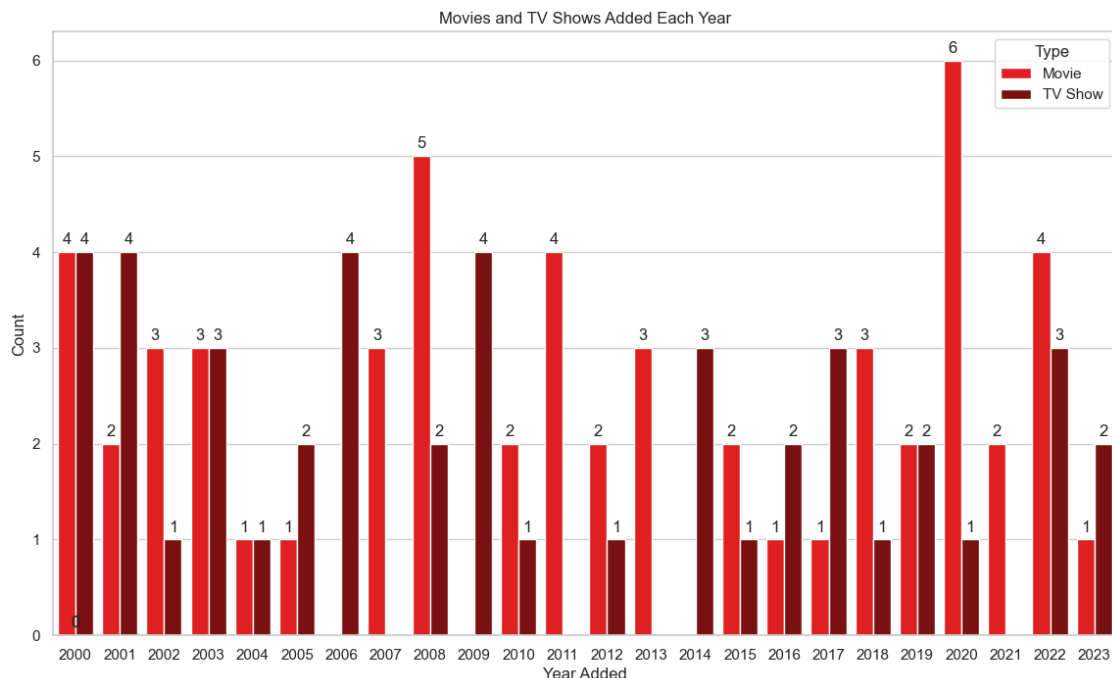


```
[114]: # Sample data creation for demonstration purposes
data = {
    'type': np.random.choice(['Movie', 'TV Show'], size=100),
```

```

    'year_added': np.random.randint(2000, 2024, size=100)
}
df1 = pd.DataFrame(data)
# Grouping the data
grouped_data = df1.groupby(['type', 'year_added']).size().
    ↪reset_index(name='count')
# Creating the plot
plt.figure(figsize=(14, 8))
sns.set_theme(style="whitegrid")
bar_plot = sns.barplot(data=grouped_data, x='year_added', y='count',
    ↪hue='type',palette=['red', 'darkred'])
# Adding data labels
for p in bar_plot.patches:
    bar_plot.annotate(format(p.get_height(), '.0f'),(p.get_x() + p.get_width() /
    ↪2., p.get_height()),
                        ha='center', va='center',xytext=(0, 9),
    ↪textcoords='offset points')
# Customizing the plot
plt.xlabel('Year Added')
plt.ylabel('Count')
plt.title('Movies and TV Shows Added Each Year')
plt.legend(title='Type')
plt.show()

```



```

[115]: fig, ax = plt.subplots(figsize=(10,6))

# Colors for Movies (Red) & TV Shows (Black)
color = ['#b20710', '#221f1f']

# Plot the data for each type (Movies & TV Shows)
for i, type_ in enumerate(df['type'].unique()):
    temp_df = df.loc[df['type'] == type_, 'year_added'].value_counts().
    ↪sort_index()
    ax.plot(temp_df.index, temp_df.values, color=color[i], label=type_,
    ↪linewidth=2)
    ax.fill_between(temp_df.index, 0, temp_df.values, color=color[i])

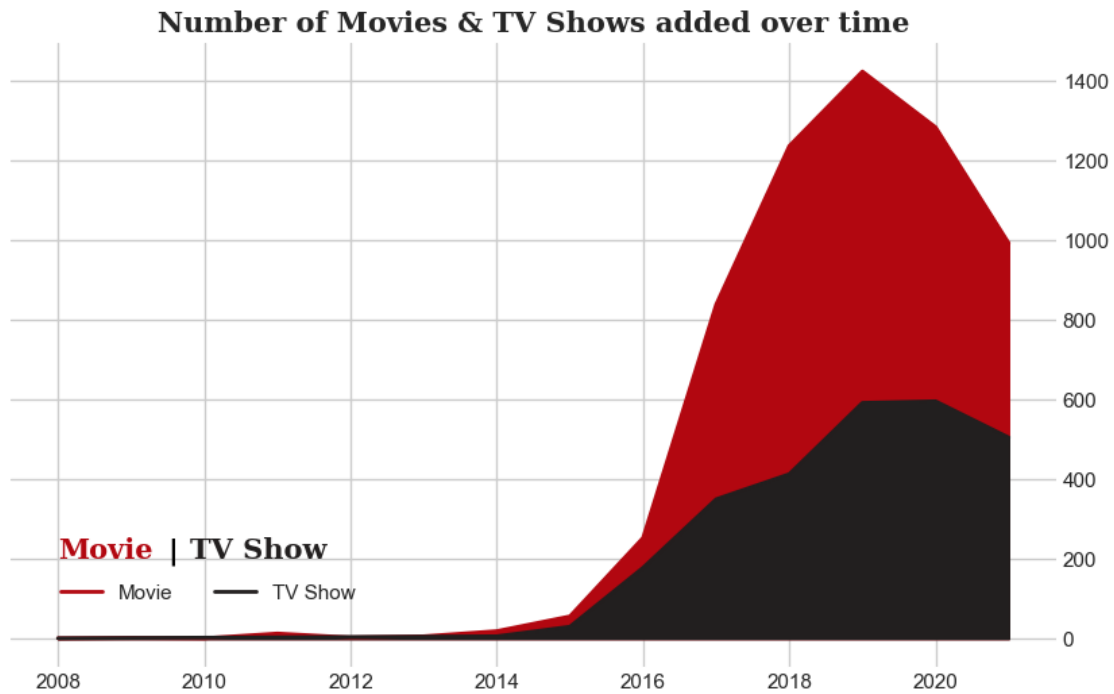
# Formatting Axes
ax.yaxis.tick_right()
ax.tick_params(axis='both', length=0)

for s in ['top', 'left', 'bottom', 'right']:
    ax.spines[s].set_visible(False)

# Title & Labels
ax.set_title('Number of Movies & TV Shows added over time', fontdict={'font':
    ↪'serif', 'size': 15, 'weight': 'bold'})

# Adding Text Labels for Legends
ax.text(2008, 200, "Movie", fontweight="bold", fontfamily='serif', fontsize=15,
    ↪color='#b20710')
ax.text(2009.5, 200, "|", fontweight="bold", fontfamily='serif', fontsize=15,
    ↪color="black")
ax.text(2009.8, 200, "TV Show", fontweight="bold", fontfamily='serif',
    ↪fontsize=15, color="#221f1f")
plt.legend(loc=(0.04, 0.09), ncol=2, frameon=False)
plt.show()

```



Insights:

1. From 2008 to 2014, Netflix saw gradual growth in the number of movies and TV shows added.
2. The year 2015 was a turning point for Netflix, with a significant spike in content additions
3. In 2020, the rate of new content additions dropped, likely due to COVID-19 production shutdowns.
4. Global Expansion in 2016 was a game changer for Netflix

```
[116]: df["director"].value_counts()[:10]
```

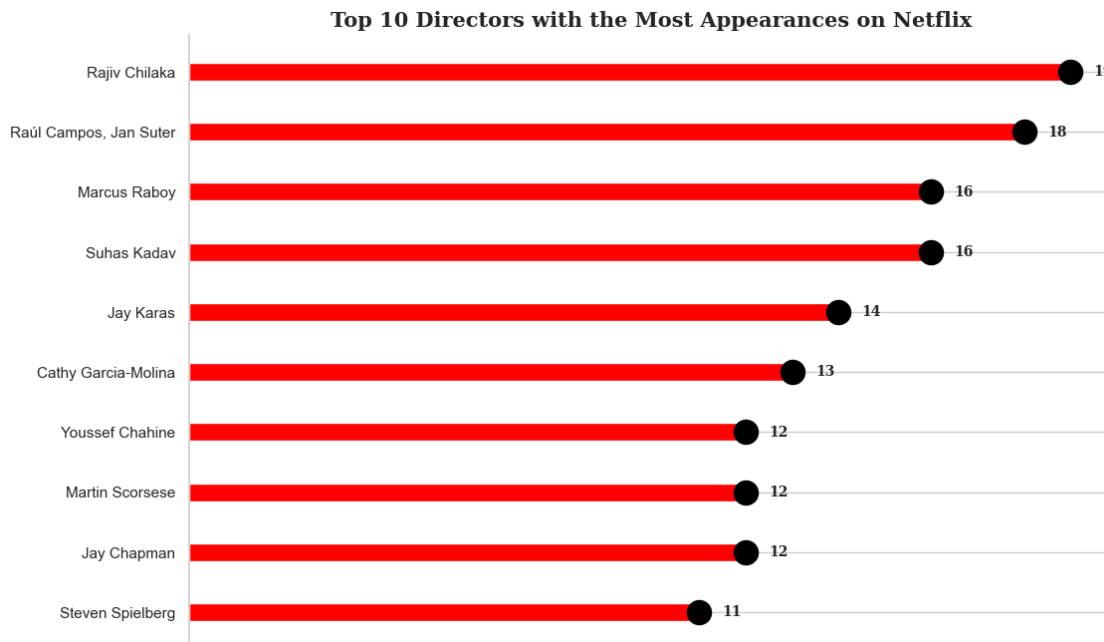
```
[116]: director
Unknown Director      2634
Rajiv Chilaka         19
Raúl Campos, Jan Suter  18
Suhas Kadav           16
Marcus Raboy          16
Jay Karas             14
Cathy Garcia-Molina   13
Martin Scorsese       12
Youssef Chahine       12
Jay Chapman           12
Name: count, dtype: int64
```

```
[117]: d_cnt = df.groupby('director')['title'].nunique().sort_values(ascending =
↪False)[0:11].reset_index()
d_cnt
```

```
[117]:
```

	director	title
0	Unknown Director	2634
1	Rajiv Chilaka	19
2	Raúl Campos, Jan Suter	18
3	Marcus Raboy	16
4	Suhas Kadav	16
5	Jay Karas	14
6	Cathy Garcia-Molina	13
7	Youssef Chahine	12
8	Martin Scorsese	12
9	Jay Chapman	12
10	Steven Spielberg	11

```
[118]: d_cnt = d_cnt.iloc[-1:-11:-1]
fig,ax = plt.subplots(figsize = (12,8))
ax.set_xticks([])
ax.barh(y = d_cnt['director'],width = d_cnt['title'],height = 0.3,color = 'red')
ax.scatter(y = d_cnt['director'], x = d_cnt['title'] , s = 300 , color = 'black')
↪)
for y,x in zip(d_cnt['director'],d_cnt['title']):
    ax.text( x + 0.5 , y , x,{ 'font':'serif', 'size':10,'weight':
↪'bold'},va='center')
for s in ['top','bottom','right']:
    ax.spines[s].set_visible(False)
    ax.set_title('Top 10 Directors with the Most Appearances on Netflix',
        {'font':'serif', 'size':15,'weight':'bold'})
plt.show()
```



Rajiv Chilaka is the most featured director on Netflix. He directed 19 movies/TV shows

1.2 Netflix Global Streaming sensation

Lets Compare top 10 countries that produce the movies and TV shows in Netflix

```
[119]: df_movies=df[df["type"]=="Movie"]
df_movies=df_movies.groupby('country')['title'].nunique().
    ↳sort_values(ascending=False)[:11].reset_index()
df_movies.drop(3,inplace=True)
df_movies["country"]=df_movies["country"].replace({"United States":
    ↳"USA","United Kingdom":"UK"})
df_movies
```

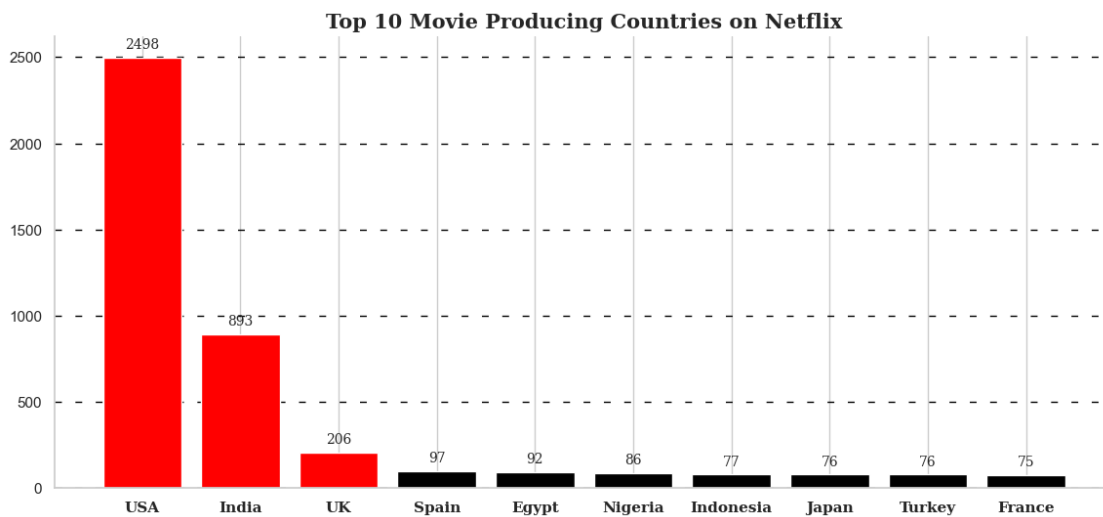
```
[119]:
```

	country	title
0	USA	2498
1	India	893
2	UK	206
4	Spain	97
5	Egypt	92
6	Nigeria	86
7	Indonesia	77
8	Japan	76
9	Turkey	76
10	France	75

```
[120]: fig,ax = plt.subplots(figsize = (14,6))
color_map = ['black' for i in range(10)]
color_map[0] = color_map[1] = color_map[2] = 'red' # highlight color
ax.bar(df_movies['country'],df_movies['title'],color = color_map,zorder = 2)

for i in df_movies.index:
    ax.text(df_movies.loc[i,'country'],df_movies.loc[i,'title'] + 75, df_movies.
        loc[i,'title'],
            {'font':'serif', 'size':10},ha = 'center',va = 'center')

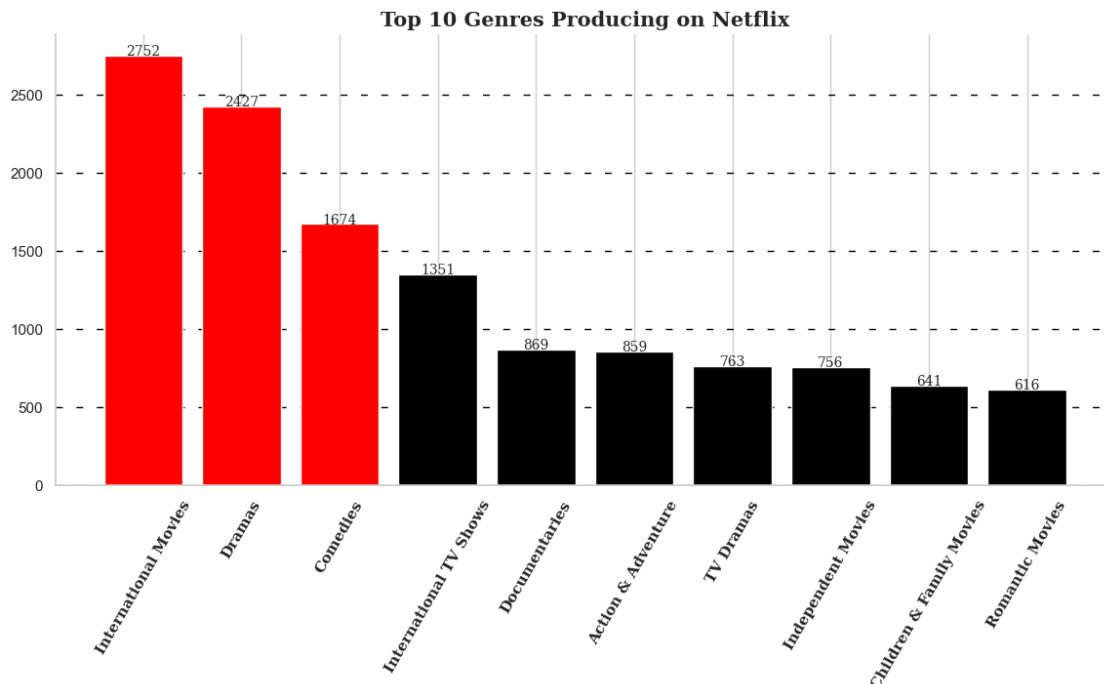
for s in ['top','left','right']:
    ax.spines[s].set_visible(False)
    ax.grid(color = 'black',linestyle = '--',axis = 'y',zorder = 0,dashes = (
        5,10))
    ax.set_xticklabels(df_movies['country'],fontweight = (
        'bold',fontfamily='serif'))
    ax.set_title('Top 10 Movie Producing Countries on Netflix',
        {'font':'serif', 'size':15,'weight':'bold'})
plt.show()
```



Insights:

1. USA produces the most content on Netflix.
2. India ranks 2nd for movies, while UK ranks 2nd for TV shows.
3. Both movies and TV shows are highly popular in the USA.
4. India is more movie-focused, as most people prefer watching movies.
5. In the UK, people enjoy both movies and TV shows equally.
6. Analysing the best week and month to add movies and TV shows on Netflix can help optimize content release.

```
[121]: df4 = copy.deepcopy(df)
df4["listed_in"] = df4["listed_in"].str.split(", ")
df4 = df4.explode("listed_in")
# Group by genre and count the number of titles in each
df_genre = df4.groupby('listed_in').size().reset_index(name='title')
df_genre = df_genre.sort_values(by='title', ascending=False).head(10)
fig, ax = plt.subplots(figsize=(14, 6))
color_map = ['black' for _ in range(10)]
color_map[0] = color_map[1] = color_map[2] = 'red' # highlight color
ax.bar(df_genre['listed_in'], df_genre['title'], color=color_map, zorder=2)
for i in df_genre.index:
    ax.text(df_genre.loc[i, 'listed_in'], df_genre.loc[i, 'title'] + 25,
            df_genre.loc[i, 'title'], {'font': 'serif', 'size': 10},
            ha='center', va='center')
ax.grid(color='black', linestyle='--', axis='y', zorder=0, dashes=(5, 10))
ax.set_xticklabels(df_genre['listed_in'], fontweight='bold',
fontfamily='serif')
plt.xticks(rotation=60)
for s in ['top', 'left', 'right']:
    ax.spines[s].set_visible(False)
ax.set_title('Top 10 Genres Producing on Netflix', {'font': 'serif', 'size':
15, 'weight': 'bold'})
plt.show()
```



Insights: 1.Top Genres Lead – The top 3 genres have the highest number of titles, showing Netflix’s strong focus on them.

2.Wide Genre Variety – Despite the top genres dominating, Netflix still offers a mix of genres to attract different audiences.

3.Smart Content Strategy – Netflix carefully distributes content across genres to reach both niche and mainstream viewers.

```
[ ]: %pip install wordcloud
```

```
[123]: df_gen=df4.groupby(['country','listed_in'])['title'].nunique().  
        ↪sort_values(ascending=False).reset_index()  
df_ind=df_gen[df_gen["country"]=="India"]
```

```
[124]: df_ind=df_ind.iloc[:10,:].reset_index(drop=True)  
df_ind
```

```
[124]:
```

	country	listed_in	title
0	India	International Movies	817
1	India	Dramas	620
2	India	Comedies	308
3	India	Independent Movies	150
4	India	Action & Adventure	127
5	India	Romantic Movies	118
6	India	Music & Musicals	88
7	India	Thrillers	88
8	India	International TV Shows	65
9	India	Horror Movies	32

```
[125]: fig, ax = plt.subplots(figsize=(14,6))  
  
# Color map for highlighting the top 3 genres  
color_map = ['black' for _ in range(10)]  
color_map[:3] = ['red', 'red', 'red'] # First 3 bars in red  
  
# Create bar chart  
ax.bar(df_ind['listed_in'], df_ind['title'], color=color_map, zorder=2)  
  
# Add labels on bars  
for i in df_ind.index:  
    ax.text(df_ind.loc[i, 'listed_in'], df_ind.loc[i, 'title'] + 25, str(df_ind.  
        ↪loc[i, 'title']),  
            fontdict={'family': 'serif', 'size': 10}, ha='center', va='center')  
  
# Grid styling  
ax.grid(color='black', linestyle='--', axis='y', zorder=0, dashes=(5,10))
```

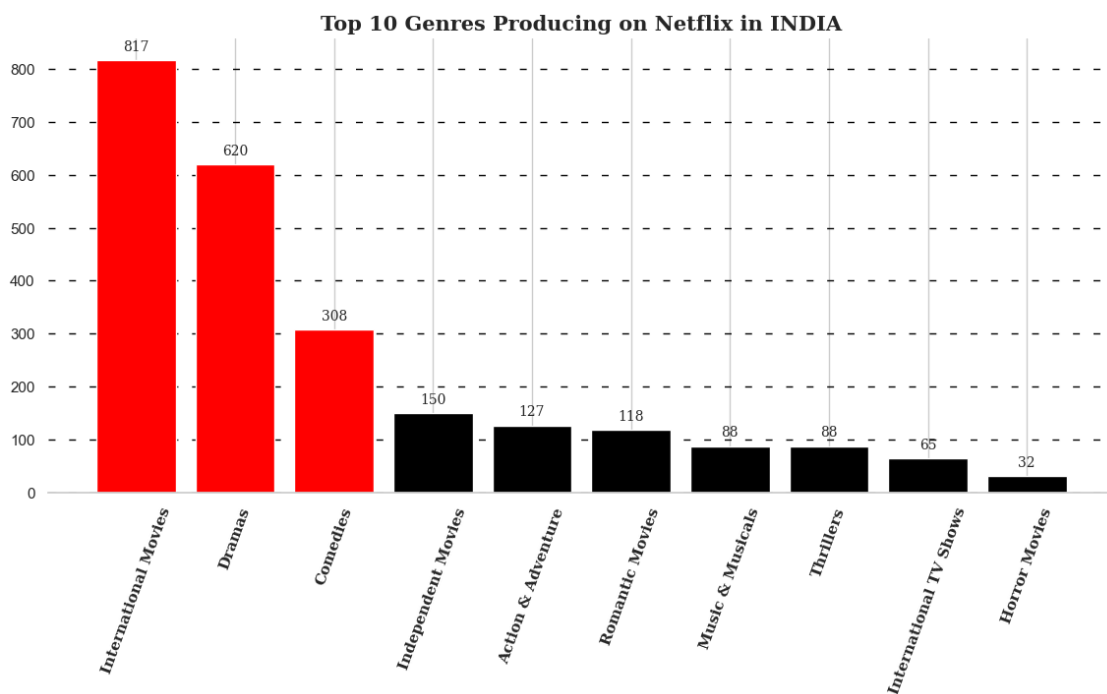
```

# Remove unnecessary spines
for s in ['top', 'left', 'right']:
    ax.spines[s].set_visible(False)

# Set x-tick labels
ax.set_xticklabels(df_ind['listed_in'], fontweight='bold', fontfamily='serif')
plt.xticks(rotation=70)

# Title
ax.set_title('Top 10 Genres Producing on Netflix in INDIA', fontdict={'family': 'serif', 'size': 15, 'weight': 'bold'})
plt.show()

```



Insights: 1. Netflix is focusing on International Movies, Dramas, and Comedies in India.

2. A total of 817 International Movies have been produced.

3. In India, Netflix invests in International Movies, Dramas, Comedies, Independent Movies, Action & Adventure, Romance, Thrillers, Music & Musicals, International TV Shows, and Horror Movies.

1.2.1 Movies/TV Shows realese count according to Month

```
[126]: df_rate=df.groupby(["month_added","type"]).agg({"type":"count"})
month=df_rate.rename(columns={"type":"count"})
month.reset_index(inplace=True)
```

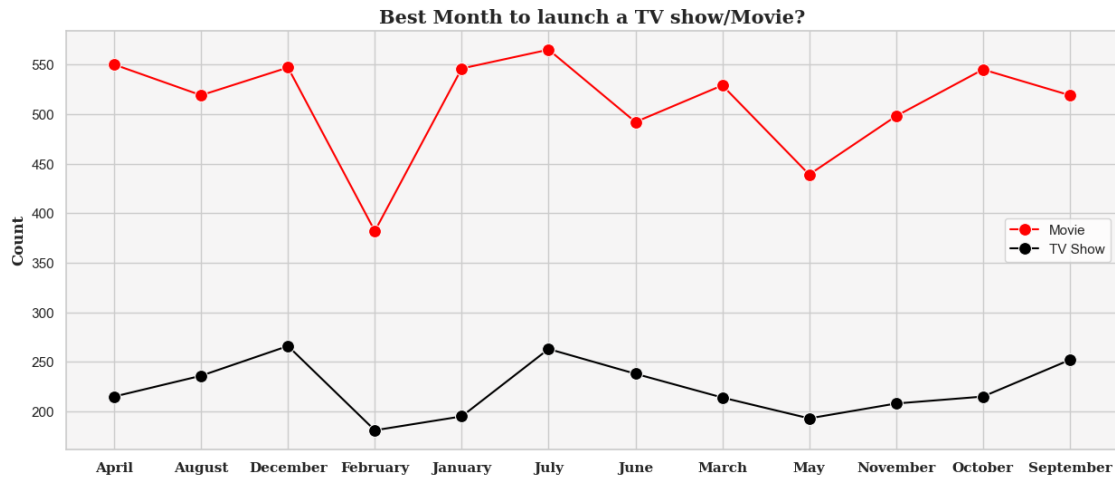
```
[127]: month
```

```
[127]:
```

	month_added	type	count
0	April	Movie	550
1	April	TV Show	215
2	August	Movie	519
3	August	TV Show	236
4	December	Movie	547
5	December	TV Show	266
6	February	Movie	382
7	February	TV Show	181
8	January	Movie	546
9	January	TV Show	195
10	July	Movie	565
11	July	TV Show	263
12	June	Movie	492
13	June	TV Show	238
14	March	Movie	529
15	March	TV Show	214
16	May	Movie	439
17	May	TV Show	193
18	November	Movie	498
19	November	TV Show	208
20	October	Movie	545
21	October	TV Show	215
22	September	Movie	519
23	September	TV Show	252

```
[128]: month_movie = month.loc[month['type'] == 'Movie'].sort_values(by = 'month_added')
month_tv = month.loc[month['type'] == 'TV Show'].sort_values(by = 'month_added')
fig,ax = plt.subplots(figsize = (15,6))
ax.set_facecolor('#f6f5f5')
sns.lineplot(data = month_movie, x = 'month_added', y = 'count',marker = 'o',
             markersize = 10,color = 'red',label = 'Movie',ax = ax)
sns.lineplot(data = month_tv, x = 'month_added', y = 'count',marker = 'o',
             markersize = 10,color = 'black',label = 'TV Show', ax = ax)
ax.set_xticklabels(month_movie['month_added'],fontweight='bold',fontfamily='serif')
plt.legend(loc = 'center right')
plt.xlabel(None)
plt.ylabel('Count',fontweight = 'bold',fontfamily='serif',fontsize = 12)
```

```
ax.set_title('Best Month to launch a TV show/Movie?',{'font':'serif', 'size':
↪15,'weight':'bold'})
plt.show()
```



1. Netflix made the most TV shows in December and the least in February.
2. No big difference in TV shows made in March, May, November, and October.
3. Netflix produced the most movies in July and the least in February.
4. April, August, and December had almost the same number of movies.

1.2.2 Movies/TV Shows realese count according to Week

```
[129]: df_rate=df.groupby(["week_added","type"]).agg({"type":"count"})
week=df_rate.rename(columns={"type":"count"})
week.reset_index(inplace=True)
```

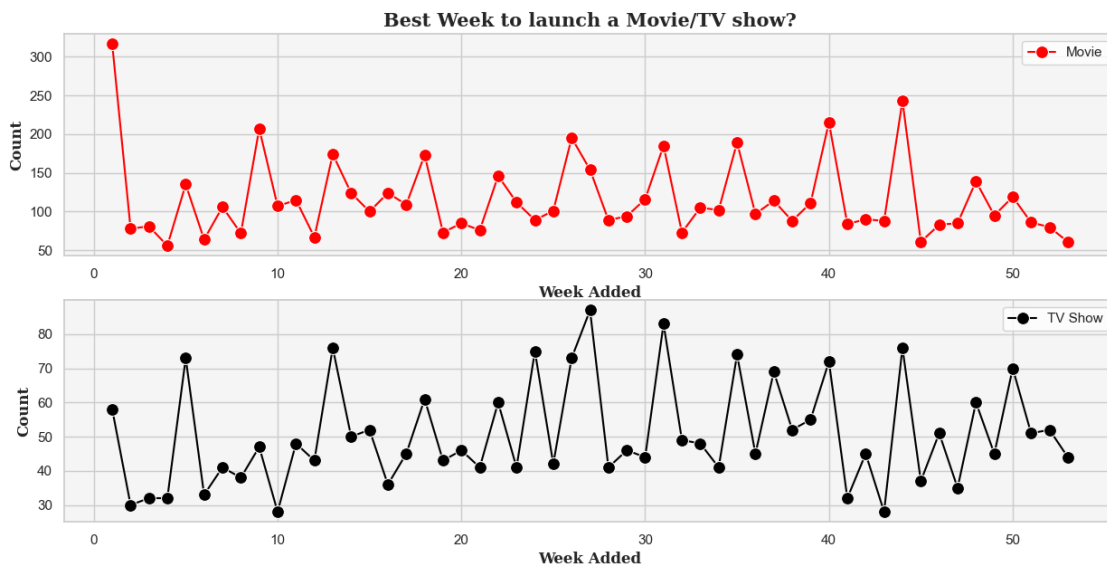
```
[130]: week
```

```
[130]:
```

	week_added	type	count
0	1	Movie	316
1	1	TV Show	58
2	2	Movie	78
3	2	TV Show	30
4	3	Movie	81
..	...	...	...
101	51	TV Show	51
102	52	Movie	80
103	52	TV Show	52
104	53	Movie	61
105	53	TV Show	44

[106 rows x 3 columns]

```
[131]: week_movie = week.loc[week['type'] == 'Movie'].sort_values(by = 'week_added')
week_tv = week.loc[week['type'] == 'TV Show'].sort_values(by = 'week_added')
fig = plt.figure(figsize = (15,7))
gs = fig.add_gridspec(2,1)
ax0 = fig.add_subplot(gs[0,:])
ax0.set_facecolor('#f6f5f5')
sns.lineplot(data = week_movie, x = 'week_added', y = 'count',marker = 'o',
             markersize = 10,color = 'red',label = 'Movie',ax = ax0)
ax0.set_xlabel('Week Added',fontweight = 'bold',fontfamily='serif')
ax0.set_ylabel('Count',fontweight = 'bold',fontfamily='serif')
ax1 = fig.add_subplot(gs[1,:])
ax1.set_facecolor('#f6f5f5')
sns.lineplot(data = week_tv, x = 'week_added', y = 'count',marker = 'o',
             markersize = 10,color = 'black',label = 'TV Show', ax = ax1)
ax1.set_xlabel('Week Added',fontweight = 'bold',fontfamily='serif')
ax1.set_ylabel('Count',fontweight = 'bold',fontfamily='serif')
ax0.set_title('Best Week to launch a Movie/TV show?',
             {'font':'serif', 'size':15,'weight':'bold'})
plt.show()
```



1.2.3 Ratings

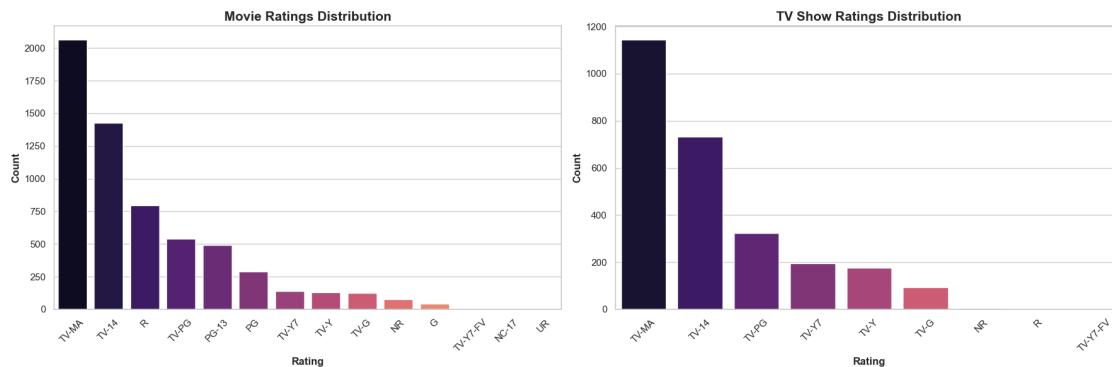
```
[132]: movie_rating = df.loc[df['type'] == 'Movie', 'rating'].value_counts().  
        ↪reset_index()  
        tv_rating = df.loc[df['type'] == 'TV Show', 'rating'].value_counts().  
        ↪reset_index()  
        print(movie_rating)  
        print(tv_rating)
```

	rating	count
0	TV-MA	2065
1	TV-14	1427
2	R	797
3	TV-PG	541
4	PG-13	491
5	PG	287
6	TV-Y7	139
7	TV-Y	131
8	TV-G	126
9	NR	75
10	G	41
11	TV-Y7-FV	5
12	NC-17	3
13	UR	3

	rating	count
0	TV-MA	1145
1	TV-14	733
2	TV-PG	324
3	TV-Y7	195
4	TV-Y	177
5	TV-G	94
6	NR	5
7	R	2
8	TV-Y7-FV	1

```
[133]: fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(18, 6))  
        # Movie Ratings  
        sns.barplot(data=movie_rating, x='rating', y='count', ax=ax1, palette='magma')  
        ax1.set_xlabel('Rating', fontsize=12, fontweight='bold')  
        ax1.set_ylabel('Count', fontsize=12, fontweight='bold')  
        ax1.set_title('Movie Ratings Distribution', fontsize=15, fontweight='bold')  
        ax1.tick_params(axis='x', rotation=45)  
  
        # TV Show Ratings  
        sns.barplot(data=tv_rating, x='rating', y='count', ax=ax2, palette='magma')  
        ax2.set_xlabel('Rating', fontsize=12, fontweight='bold')  
        ax2.set_ylabel('Count', fontsize=12, fontweight='bold')  
        ax2.set_title('TV Show Ratings Distribution', fontsize=15, fontweight='bold')
```

```
ax2.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()
```



1.2.4 Target Audience Distribution for Movies and TV Show

```
[134]: # Define age group function
def age_group(x):
    if x in ['TV-Y', 'TV-G', 'G']:
        return 'Little Kids'
    elif x in ['TV-Y7', 'TV-Y7-FV', 'TV-PG', 'PG']:
        return 'Older Kids'
    elif x in ['PG-13', 'TV-14']:
        return 'Teens'
    else:
        return 'Adults'

    # Apply age group function to movie_rating and tv_rating
movie_rating['age'] = movie_rating['rating'].apply(lambda x: age_group(x))
age_m_rating = movie_rating.groupby('age')['count'].sum().
    ↪sort_values(ascending=False)
tv_rating['age'] = tv_rating['rating'].apply(lambda x: age_group(x))
age_t_rating = tv_rating.groupby('age')['count'].sum().
    ↪sort_values(ascending=False)
age_m_rating
```

```
[134]: age
Adults      2943
Teens       1918
Older Kids   972
Little Kids  298
Name: count, dtype: int64
```

```
[135]: import matplotlib.pyplot as plt

# Sample Data (Replace with actual data)
age_m_rating = {
    'age_group': ['Adults', 'Teens', 'Older Kids', 'Little Kids'],
    'count': [48, 31.3, 15.9, 4.9]
}
age_t_rating = {
    'age_group': ['Adults', 'Teens', 'Older Kids', 'Little Kids'],
    'count': [43, 27.4, 19.4, 10.1]
}

# Define different color palettes for Movies & TV Shows
movie_colors = ['#D32F2F', '#E57373', '#FFCDD2', '#F8BBD0']
tvshow_colors = ['#1B5E20', '#4CAF50', '#A5D6A7', '#C8E6C9']

# Create the figure and gridspec layout
fig = plt.figure(figsize=(15, 8))
gs = fig.add_gridspec(1, 2)

# First subplot for Movies
ax0 = fig.add_subplot(gs[0, 0])
wedges, texts, autotexts = ax0.pie(
    age_m_rating['count'],
    labels=age_m_rating['age_group'],
    colors=movie_colors, explode=[0.1, 0, 0, 0],
    autopct='%1.1f%%', wedgeprops={'linewidth': 2, 'edgecolor': 'white'},
    textprops={'fontsize': 13, 'color': 'black'}
)

# Add numbers to the center of each pie slice
for autotext in autotexts:
    autotext.set_color('black')
    autotext.set_fontsize(12)
ax0.set_title('Movies Vs Audience Age Group', fontdict={'family': 'serif', '
    ↪ 'fontsize': 15, 'fontweight': 'bold'})

# Second subplot for TV Shows
ax1 = fig.add_subplot(gs[0, 1])
wedges, texts, autotexts = ax1.pie(
    age_t_rating['count'],
    labels=age_t_rating['age_group'],
    colors=tvshow_colors, explode=[0.1, 0, 0, 0],
    autopct='%1.1f%%', wedgeprops={'linewidth': 2, 'edgecolor': 'white'},
    textprops={'fontsize': 13, 'color': 'black'}
)
```

```

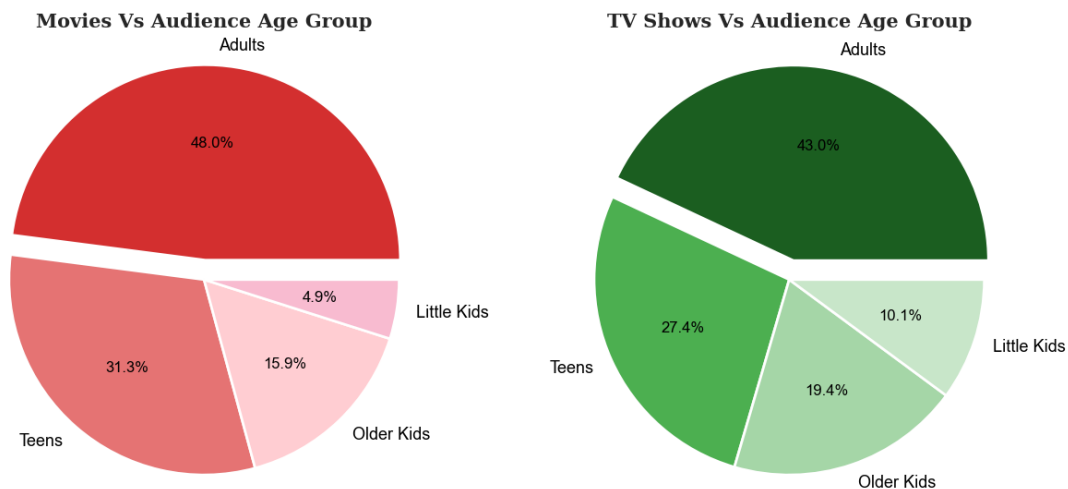
# Add numbers to the center of each pie slice
for autotext in autotexts:
    autotext.set_color('black')
    autotext.set_fontsize(12)
ax1.set_title('TV Shows Vs Audience Age Group', fontdict={'family': 'serif',
    ↪ 'fontsize': 15, 'fontweight': 'bold'})

# Add main title
fig.suptitle('Netflix Target Audience Analysis', fontdict={'family': 'serif',
    ↪ 'fontsize': 18, 'fontweight': 'bold'})

# Show the plot
plt.show()

```

Netflix Target Audience Analysis



1. Movies • Roughly half of the films available on the platform cater to adult audiences, while around 30% of the overall content is designed for teenagers, and the remaining 20% is tailored for children

2. TV Shows • The pattern holds true for television shows as well. The only distinction is that 30% of the content is targeted towards children, which signifies the inclusion of anime shows within Netflix's offerings.

1.2.5 Netflix Movie and TV Show runtime

- Analysing the Run time for Movies and TV Shows

```
[136]: tv_duration = df.loc[df['type'] == 'TV Show', 'duration'].value_counts().
        ↪reset_index()
tv_duration.columns = ['duration', 'count'] # Rename columns properly

# Replace season labels
season_mapping = {
    '1 Season': '1S', '2 Seasons': '2S to 3S', '3 Seasons': '2S to 3S',
    '4 Seasons': '4S to 6S', '5 Seasons': '4S to 6S', '6 Seasons': '4S to 6S',
    '7 Seasons': '7S to 9S', '8 Seasons': '7S to 9S', '9 Seasons': '7S to 9S',
    '10 Seasons': '10S to 17S', '11 Seasons': '10S to 17S',
    '12 Seasons': '10S to 17S', '13 Seasons': '10S to 17S',
    '14 Seasons': '10S to 17S', '15 Seasons': '10S to 17S',
    '16 Seasons': '10S to 17S', '17 Seasons': '10S to 17S'}

tv_duration['duration'] = tv_duration['duration'].replace(season_mapping)

# Group by new duration categories
tv_duration = tv_duration.groupby('duration')['count'].sum().
        ↪sort_values(ascending=False).reset_index()

# Rename columns
tv_duration.rename(columns={'duration': 'Season', 'count': 'Count'},
        ↪inplace=True)

# Display result
tv_duration
```

```
[136]:
```

	Season	Count
0	1S	1793
1	2S to 3S	624
2	4S to 6S	193
3	7S to 9S	49
4	10S to 17S	17

```
[137]: # Filter for Movies and count durations
movie_duration = df.loc[df['type'] == 'Movie', 'duration'].value_counts().
        ↪reset_index()

# Rename columns properly
movie_duration.columns = ['duration', 'count']

# Remove ' min' from duration and convert to int
movie_duration['duration'] = movie_duration['duration'].str.replace(' min', '',
        ↪regex=True).astype(int)

# Define bins for categorizing film lengths
bin_range = [0, 30, 90, 150, 210, float('inf')]
```

```

bin_labels = ['Short Films', 'Medium Films', 'Standard Films', 'Long Films',
↳ 'Epic Films']

# Categorize movies into bins
movie_duration['duration'] = pd.cut(movie_duration['duration'], bins=bin_range,
↳ labels=bin_labels)

# Group by film length category
movie_duration = movie_duration.groupby('duration')['count'].sum().
↳ sort_values(ascending=False).reset_index()

# Rename columns
movie_duration.rename(columns={'duration': 'Film Length', 'count': 'Count'},
↳ inplace=True)
movie_duration

```

```

[137]:
      Film Length  Count
0  Standard Films   3893
1   Medium Films   1863
2    Long Films    234
3   Short Films    130
4    Epic Films     11

```

```

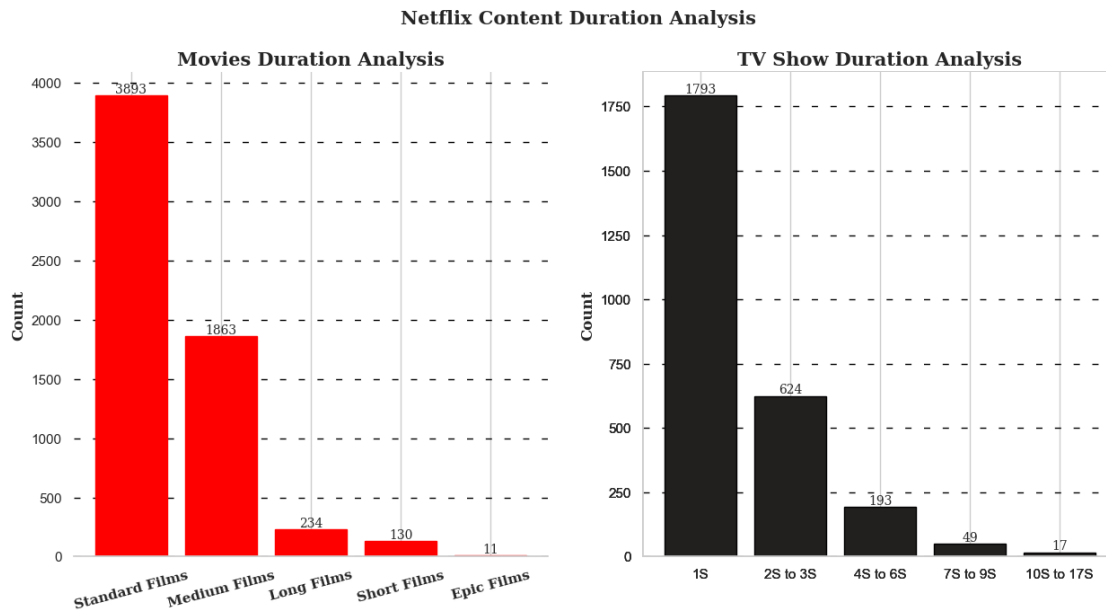
[138]: fig = plt.figure(figsize = (15,7))
gs = fig.add_gridspec(1,2)
ax0 = fig.add_subplot(gs[0,0])
color_map = ['LightCoral', 'Salmon', 'DarkSalmon', 'LightSalmon', 'Crimson']
ax0.bar(movie_duration['Film Length'], height=
↳ movie_duration['Count'], edgecolor='red', color = 'red', zorder = 2)
for i in movie_duration.index:
    ax0.text(movie_duration.loc[i, 'Film Length'], movie_duration.loc[i, 'Count']+
↳ 50, movie_duration.loc[i, 'Count'],
              {'font': 'serif', 'size': 10}, ha = 'center', va = 'center')
    ax0.grid(color = 'black', linestyle = '--', axis = 'y', zorder = 0, dashes =
↳ (5,10))
    ax0.set_xticklabels(movie_duration['Film Length'],
                        fontweight = 'bold', fontfamily='serif', rotation = 15)
    ax0.set_ylabel('Count', fontweight = 'bold', fontfamily='serif')
for s in ['top', 'left', 'right']:
    ax0.spines[s].set_visible(False)
    ax0.set_title('Movies Duration Analysis', {'font': 'serif', 'size':
↳ 15, 'weight': 'bold'})
    ax1 = fig.add_subplot(gs[0,1])
    color_map = ['lightpink', 'deeppink', 'mediumvioletred', 'darkred',
↳ 'firebrick']

```

```

ax1.bar(tv_duration['Season'],height_
↪tv_duration['Count'],edgecolor='black',color = '#221f1f',zorder = 2)
for i in tv_duration.index:
    ax1.text(tv_duration.loc[i,'Season'],tv_duration.loc[i,'Count'] +_
↪25,tv_duration.loc[i,'Count'],
              {'font':'serif', 'size':10},ha = 'center',va = 'center')
ax1.grid(color = 'black',linestyle = '--',axis = 'y',zorder = 0,dashes =_
↪(5,10))
ax1.set_ylabel('Count',fontweight = 'bold',fontfamily='serif')
for s in ['top','left','right']:
    ax1.spines[s].set_visible(False)
ax1.set_title('TV Show Duration Analysis',
              {'font':'serif', 'size':15,'weight':'bold'})
fig.suptitle('Netflix Content Duration Analysis',fontproperties = {'family':
↪'serif', 'size':15,'weight':'bold'})
plt.show()

```



1.3 Insights Based on Graphical and Non Graphical Analysis

- About 70% of Netflix's content consists of movies, while the remaining 30% is TV shows.
- Netflix started adding content in 2008, with very limited titles until 2014.
- 2015 saw a major surge in the number of movies and TV shows added, and this growth continued.
- 2019 recorded the highest number of new movies and TV shows added to Netflix.
- In 2020 and 2021, content additions dropped, likely due to the pandemic. However, the decrease in movies was more significant than in TV shows.

- Since 2018, the number of movies added has been decreasing, while the number of TV shows has been steadily increasing.
- By mid-2020, TV shows outnumbered movies, indicating their growing popularity.
- Netflix features content from 4,993 different directors worldwide.
- Netflix includes movies and TV shows from 122 countries, with the United States contributing the most (about 37% of the total content).
- The release years of shows on Netflix are mostly between 2005 and 2021.
- Netflix offers content with various ratings catering to different age groups, including Teens, Adults, Kids, and Families.
- The most common rating for movies and TV shows on Netflix is TV-MA (for mature audiences).
- Child-friendly ratings like TV-Y7, TV-Y7 FV, PG, TV-G, G, TV-Y, and TV-PG are limited in most countries, except for the United States, where they are more widely available.
- The most popular genres on Netflix for both movies and TV shows are international content, Dramas, and Comedies.
- Each country has its own preferred genres, but the United States has a wide variety covering almost all genres.
- Examples: Korean dramas are famous in Korea, British TV series are common in the UK, and Anime (both movies and series) is popular in Japan.
- Indian actors appear in the highest number of movies on Netflix, with the top five most-featured actors coming from India.
- Short films and movies with a shorter runtime have gained popularity over the last 10 years.
- Movies from India and Germany tend to be older, with an average age of 8 years.
- Movies from Spain are mostly newer, with an average age of just 1 year.
- Japanese TV shows are generally older, with an average age of 5 years. • TV shows from India and Spain are relatively new, averaging around 1 year old.

Business Insights: • Most of Netflix's content is from the year 2000 onward, with very few titles from before 2000.

- Older content is scarce, meaning senior citizens who may prefer classic films and shows are an underserved audience.
- Over 80% of Netflix content falls under the following ratings:
 - * TV-MA - For viewers 17 and older (mature audience).
 - * TV-14 - Suitable for viewers 14 and older.
 - * TV-PG, PG-13, PG - Requires parental guidance.
 - * R - Restricted, not suitable for viewers under 17.
- Most of Netflix's content is aimed at adults, while only 20% is designed for kids under 13 (often with parental supervision).

- The most popular genres on Netflix include:
 - * International Movies & TV Shows.
 - * Dramas.
 - * Comedies.
 - * Action & Adventure.
 - * Children & Family Movies.
 - * Thrillers.
- 75% of Netflix's content comes from just 10 countries, while the remaining 25% comes from other regions.
- Expanding content from more countries could help Netflix reach a broader audience.
- Shorter content is becoming more popular, including:
 - * Movies lasting between 75 to 150 minutes.
 - * TV shows with 1 to 3 seasons.
 - * This trend should be considered for future productions.
- A decline in content additions was observed in 2020 and 2021 across all countries and content types, likely due to the pandemic.

Recommendations • Limited genre diversity in most countries, except the U.S., suggests that Netflix's current genres are better suited for American audiences.

- Many countries need more content tailored to their local preferences.
- Example: Indian mythological content is very popular and could be expanded.
- Creating regional genres may also attract global audiences, similar to how Japanese Anime has gained worldwide popularity.
- Country-specific content strategy is essential.
- Content should be designed to match the interests and demographics of each country.
- Netflix should increase content in specific ratings based on audience demand in each region.

Example:

- India, a country with a huge population, mostly has content rated TV-MA, TV-14, and TV-PG.
- This limits options for younger viewers (under 14) and older audiences (above 35).
- Producing content for these neglected age groups could expand Netflix's reach in India.